LoGST: A Coordinate-Invariant Local-Global Spatial Transformer for Gene Expression Prediction from Histopathology

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Abstract

Spatial transcriptomics maps gene expression across tissue, but remains expensive and experimentally demanding. Predicting spatial gene expression directly from routine hematoxylin-and-eosin (H&E) histology images is a lower-cost alternative, yet most existing methods are trained and evaluated within a single tissue type, leaving cross-organ generalization untested. We introduce LoGST, a graph transformer that predicts gene expression at every tissue spot from H&E patch features in a single forward pass, without requiring any expression measurements at test time.

LoGST builds a spatial graph over tissue spots and encodes their geometry through pairwise distances only — never through absolute coordinates — making predictions invariant to slide translation, rotation, and reflection. Three complementary attention streams capture context at different scales: distance-biased local graph attention over neighboring spots captures tissue microstructure, global self-attention over all spots provides slide-wide context, and a learned slide-summary token aggregates holistic information that is broadcast back to every spot.

We evaluate on HEST-1k in three settings with matched gene panels, training budgets, and frozen pathology features (UNI+CONCH): within-organ prediction, pooled cross-sample validation, and leave-one-organ-out transfer to an unseen organ type. Against five baseline methods, LoGST achieves the best average within-organ Pearson correlation — a 6.6% relative improvement over the strongest baseline — while remaining competitive under cross-organ transfer. With 4.88M parameters and single-pass inference, LoGST offers an efficient and generalizable approach to spatial gene-expression prediction across diverse tissue types.

1. Introduction

Spatial transcriptomics (ST) measures gene expression together with its location in tissue, making it possible to study spatial heterogeneity, tissue organization, and interactions between cells [9]. Despite its value, ST remains costly and experimentally demanding. Hematoxylin-and-eosin (H&E) whole-slide images are routinely collected and contain morphological patterns associated with tissue composition and molecular activity. This has motivated computational methods that predict spatial gene expression from histology, potentially extending molecular analysis to samples for which ST measurements are unavailable.

Most histology-to-expression methods are developed and evaluated separately within individual tissue cohorts. In this setting, training and test slides share an organ type and often similar collection characteristics. Strong within-cohort performance therefore does not establish that a model will generalize across organs. Such transfer is challenging because organs differ in tissue architecture, cell composition, expression distributions, staining, and acquisition conditions. We address this gap with two multi-organ evaluations: pooled cross-validation tests held-out samples from organ types represented during training, whereas leave-one-organ-out (LOOO) evaluation tests an entirely unseen organ (Fig. 1). Together with standard intra-cohort evaluation, these settings separate within-organ prediction, generalization to new samples from known organs, and transfer to a new organ.

Existing methods capture spatial context using graph networks, large attention models, generative objectives, or explicit count-distribution priors. These approaches can be effective, but their additional complexity makes it harder to determine which spatial signals are necessary for accurate and transferable prediction. We instead ask whether a compact direct-regression model can provide a strong baseline when it is given a suitable geometric inductive bias. Our central design choice is to use coordinates only through pairwise Euclidean distances. With histology features held fixed, this makes the spatial pathway unchanged by translations, rotations, and reflections of the coordinate system, without requiring frame averaging.

We introduce LoGST (Local-Global Spatial Transformer), a single-pass model that predicts a panel of genes at every spot in a slide. Each layer combines three complementary forms of context: distance-aware local attention captures relationships between nearby spots, global self-attention models slide-wide dependencies, and an attention-pooled slide representation summarizes overall morphology. A linear prediction head then produces expression for all spots and genes in one forward pass, without generative sampling. This design is compact, coordinate-invariant in its spatial pathway, and directly testable through component ablations.

Research questions. We organize the study around five questions:

- RQ1 Can direct regression compete under standard within-cohort evaluation?
- RQ2 Does LoGST generalize across organs without per-organ training?
- RQ3 Which spatial context components drive performance?
- RQ4 Do predictions recover biological spatial structure beyond per-gene correlation?
- RQ5 How efficient and scalable is the model?

Contributions. (1) We propose a compact spatial transformer that combines local, global, and slide-level context while using only pairwise distances for spatial encoding. (2) We establish a feature-matched evaluation protocol with identical frozen UNI features, data splits, gene panels, training budgets, and nested validation across intra-cohort, POOLED, and LOOO settings on HEST-1k. Under intra-cohort evaluation, LoGST improves average Pearson correlation by 6.6% relative to the strongest of five controlled baselines and remains competitive in both multi-organ settings. (3) We analyze the model through factorial context and objective ablations, spatial domain recovery, representation and expression-map visualizations, numerical invariance checks, and measured inference scaling.

2. Related Work

2.1. Histology-based gene expression prediction

Histology-to-expression methods differ mainly in how they represent morphology and exchange information between spatial locations. ST-Net [3] predicts expression from individual image patches with a convolutional network, providing a strong spot-level formulation but no explicit interaction between spots. HisToGene [6] introduces a transformer to model dependencies across a slide, while Hist2ST [13] combines transformer layers with a spatial graph. BLEEP [12] instead aligns histology and expression embeddings through contrastive learning and predicts by retrieval. TRIPLEX [2] integrates spot, neighbourhood, and global image information at multiple resolutions. These methods demonstrate that spatial and contextual information improves upon independent patch regression, but they use substantially different features, objectives, and evaluation procedures in their original studies.

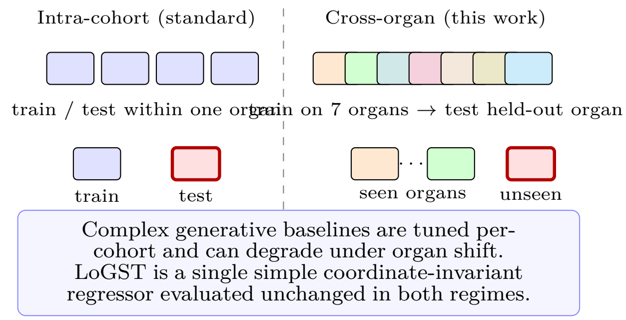

Figure 1. Two evaluation regimes. Standard benchmarks train and test within one tissue cohort (left). We additionally test the harder cross-organ transfer (right), where a model must predict expression for slides from organs unseen during training. LoGST is evaluated in both without per-cohort redesign.

Recent methods introduce additional structural assumptions. MCToGene [10] models higher-order interactions among multiple cells. Gene-DML [8] uses dual pathways and multi-level discrimination, whereas HyperST [14] represents cross-modal structure in hyperbolic space. Large pretrained models such as STPath [15] learn across organs and platforms with explicit metadata. LoGST is complementary to these approaches: it uses frozen UNI features [1], direct regression, and a compact spatial transformer so that the effects of local, global, and slide-level context can be evaluated under a common protocol.

2.2. Geometric structure in spatial models

Spatial prediction depends on the coordinate representation supplied to a model. Absolute position embeddings can encode location within a particular coordinate frame, but their output may change when an equivalent tissue layout is translated, rotated, or reflected. Geometric networks address this issue by constructing interactions from relative quantities [7]. LoGST uses an invariant construction in which coordi-

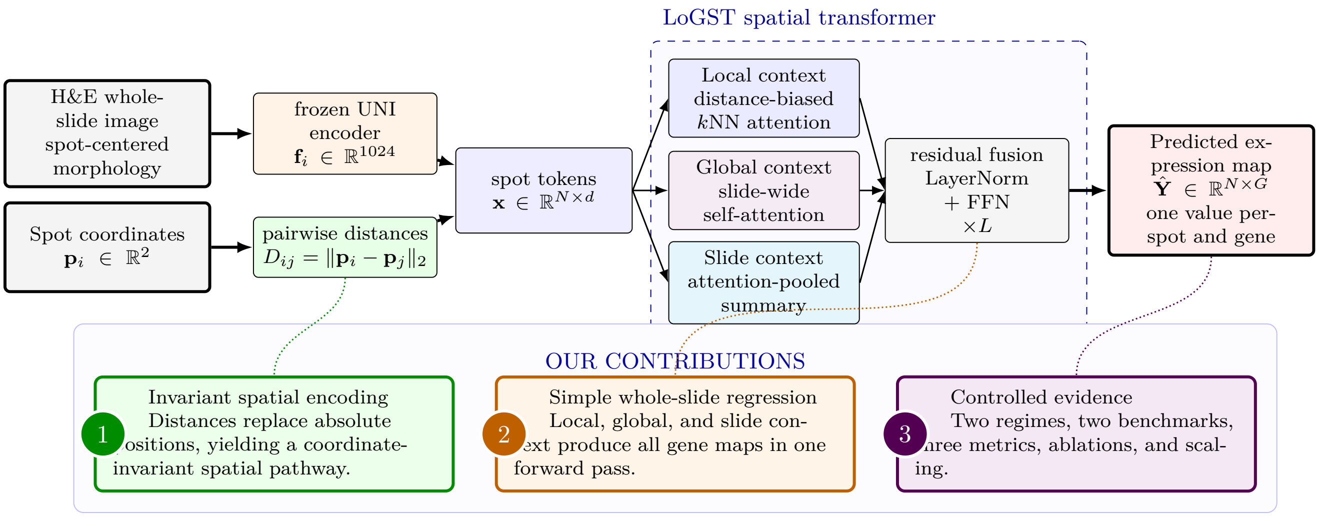

Figure 2. LoGST overview and contributions. The model receives frozen UNI morphology features and spot coordinates. Coordinates affect the network only through pairwise distances. Each layer combines local, global, and slide-level context before a linear head predicts the expression of G genes at all N spots. The design provides an invariant spatial pathway, direct single-pass inference, and a controlled framework for evaluating which forms of spatial context are useful.

nates enter its spatial pathway only through pairwise Euclidean distances. Because these distances are preserved by translations, rotations, and reflections, no transformed coordinate frames or prediction averaging are required. This guarantee applies to the spatial pathway for fixed histology features; it does not imply that the frozen image encoder is invariant to rotations of the image itself.

2.3. Generalization across samples and organs

Most published spatial gene-expression benchmarks train and test within the same tissue cohort. HEST-1k [4] provides data from multiple cohorts and organs, but its standard evaluation remains organized by cohort. Consequently, a high within-cohort score can reflect adaptation to organ-specific morphology, expression distributions, or acquisition conditions. Recent benchmarking work likewise identifies cross-sample generalization as a major challenge [5]. We evaluate two complementary multi-organ settings. POOLED cross-validation trains jointly on all organ groups and tests held-out samples from organ types seen during training. LOOO trains on seven organ groups and evaluates on the held-out eighth group, directly measuring transfer to an unseen organ. This distinction prevents performance on known organs from being interpreted as evidence of unseen-organ generalization.

3. Method

LoGST directly maps the spots of one tissue slide to their gene-expression profiles. For a slide with N spots, the input consists of frozen UNI features $\mathbf{F} = [\mathbf{f}_1, \ldots, \mathbf{f}_N]^\top \in \mathbb{R}^{N \times 1024}$ and spatial coordinates $\mathbf{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_N]^\top \in \mathbb{R}^{N \times 2}$. The output $\hat{\mathbf{Y}} \in \mathbb{R}^{N \times G}$ contains the predicted log-normalized expression of a G-gene panel at every spot. Slides are processed independently, so the model requires neither cross-slide padding nor a fixed number of spots.

3.1. Coordinate-invariant spatial encoding

We first compute the pairwise Euclidean distance matrix $D_{ij} = \|\mathbf{p}_i - \mathbf{p}_j\|_2$. For each spot i, we retain the indices $\mathcal{N}(i)$ and distances of its k nearest neighbours, excluding the spot itself. Distances are expanded into M radial-basis features,

$$\boldsymbol{\phi}(D_{ij}) = \left[\exp\!\left(-\gamma_m D_{ij}^2\right)\right]_{m=1}^{M}, \tag{1}$$

where the fixed bandwidth parameters γ_m are logarithmically spaced. These features influence attention weights but are never added to the spot embeddings as absolute positions.

This construction yields coordinate-system invariance. For any orthogonal matrix R and translation $\mathbf{t}$,

$$\|R\mathbf{p}_i + \mathbf{t} - (R\mathbf{p}_j + \mathbf{t})\|_2 = \|\mathbf{p}_i - \mathbf{p}_j\|_2. \tag{2}$$

The neighbour graph, radial-basis features, and predictions are therefore unchanged by translations, rotations, and reflections of the coordinates, up to floating-point error, when the UNI features are held fixed. The result concerns transformations of the coordinate system and does not assert rotation invariance of the frozen histology encoder.

3.2. The LoGST layer

The input features are projected to hidden width d, $\mathbf{X}^{(0)} = \mathbf{F}W_{\text{in}} + \mathbf{b}_{\text{in}}$. The same neighbour graph and radial-basis features are reused in all L transformer layers. At layer ℓ, a pre-normalized representation $\mathbf{H} = \text{LN}(\mathbf{X}^{(\ell)})$ is passed through three context pathways.

Local distance-biased attention. Each spot attends to its k neighbours with a learned, distance-dependent bias for each attention head. For head h, the attention weights are

$$\alpha_{ij}^{h} = \text{softmax}_{j\in\mathcal{N}(i)}\left(\frac{(\mathbf{q}_i^h)^\top \mathbf{k}_j^h}{\sqrt{d_h}} + b_h(\phi(D_{ij}))\right), \quad (3)$$

where $d_h = d/H$ for H heads. The local output is the concatenation of the weighted value sums, followed by a learned output projection. Morphological similarity is captured by the query–key term, while spatial distance contributes through the learned radial-basis bias.

Global self-attention. In parallel, standard multi-head self-attention [11] allows every spot to attend to all N spots. This pathway captures long-range dependencies that may not be connected in the local graph. It uses morphology features only and is permutation-equivariant with respect to spot order.

Attention-pooled slide context. A learned scalar scoring function produces $a_i = \text{softmax}_i(s(\mathbf{h}_i))$. The slide representation is

$$\mathbf{g} = W_g\left(\sum_{i=1}^{N} a_i \mathbf{h}_i\right) + \mathbf{b}_g, \quad (4)$$

and is broadcast to every spot. Unlike a learned class token, this representation is computed directly from the slide's spot features and summarizes its morphological composition.

For the complete V3 model, the local output, global output, and broadcast slide representation are summed and added to the residual stream. A second pre-normalization residual update applies a two-layer feed-forward network with expansion factor four and GELU activation:

$$\widetilde{\mathbf{X}}^{(\ell)} = \mathbf{X}^{(\ell)} + \mathbf{L}^{(\ell)} + \mathbf{A}^{(\ell)} + \mathbf{G}^{(\ell)}, \quad (5)$$

$$\mathbf{X}^{(\ell+1)} = \widetilde{\mathbf{X}}^{(\ell)} + \text{FFN}\Big(\text{LN}(\widetilde{\mathbf{X}}^{(\ell)})\Big). \quad (6)$$

After L layers, layer normalization and a linear head map each spot representation to its G predicted gene values. The model produces the complete slide prediction in one forward pass.

3.3. Training objective and staged design

For a slide with measured expression $\mathbf{Y}$, we optimize mean squared error together with a gene-wise Pearson-correlation term:

$$\mathcal{L} = \text{MSE}(\hat{\mathbf{Y}}, \mathbf{Y}) + \lambda\left(1 - \frac{1}{|\mathcal{V}|}\sum_{g\in\mathcal{V}} \rho_g\right), \quad (7)$$

where ρ_g is the Pearson correlation between predicted and measured expression across the N spots, $\mathcal{V}$ is the set of genes with nonzero target variance on that slide, and $\lambda = 0.5$. MSE constrains the scale of the predictions, while the correlation term encourages the model to reproduce spot-to-spot expression patterns.

We developed the architecture incrementally: V0 retains the projection, residual feed-forward blocks, and prediction head but removes all context pathways; V1 adds global self-attention; V2 adds local distance-biased attention; and V3 adds the pooled slide representation. V3 is the complete model used in the main experiments. To avoid attributing performance to this particular addition order, the factorial ablation independently enables or disables the local (L), global (G), and slide (S) pathways (Table 7).

4. Experiments

4.1. Datasets

We evaluate on HEST-1k [4], which provides ten tissue cohorts, each with a patient-stratified k-fold split; for presentation we aggregate the ten cohorts into eight organ groups (breast = IDC + LYMPH_IDC, colorectal = COAD + READ, others one cohort each). For every slide we predict the 50 most highly variable genes after $\log(1+x)$ normalization, using identical frozen UNI [1] features across all methods.

4.2. Experimental setup

Two regimes. We evaluate under (i) an intra-cohort protocol—the standard per-cohort k-fold cross-validation provided with the HEST-1k benchmark [4], training separately within each cohort—and (ii) a harder cross-organ protocol with two sample-disjoint settings: leave-one-organ-out (looo), where seven organ groups train and the eighth tests, and pooled cross-validation (pooled). Patient metadata are unavailable for some cohorts, so we do not claim the pooled splits are patient-disjoint.

Metrics and protocol. We report the Pearson correlation coefficient (PCC), Spearman correlation (SCC), and mean squared error (MSE) between predicted and measured expression, averaged over folds and three

Table 1. Intra-cohort gene expression prediction on HEST-1k, per cohort (Pearson ↑). Mean_{std} over three seeds with nested-validation; best per row in bold. All methods use identical frozen UNI + CONCH features (1536-dim), 50-gene panels, and the same training budget (100 epochs, patience 20, inner-validation selection). HCC, LYMPH_IDC, and READ excluded (see Appendix E).

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM ICLR'25	LoGST+CONCH
IDC Breast	0.551±0.01	0.309±0.07	0.567±0.01	0.519±0.00	0.439±0.01	**0.581±0.01**
COAD Colorectal	0.242±0.01	0.295±0.01	0.285±0.01	0.314±0.01	0.177±0.01	**0.318±0.01**
CCRCC Kidney	0.163±0.02	0.184±0.02	**0.256±0.02**	0.175±0.02	0.123±0.00	0.252±0.01
LUNG Lung	0.518±0.01	0.470±0.01	0.587±0.01	0.562±0.01	0.423±0.03	**0.589±0.01**
PAAD Pancreas	0.359±0.03	0.247±0.02	0.404±0.04	0.419±0.03	0.233±0.02	**0.437±0.03**
PRAD Prostate	0.294±0.02	0.333±0.02	0.384±0.02	0.348±0.01	0.207±0.02	**0.410±0.01**
SKCM Skin	0.665±0.01	0.522±0.03	0.653±0.01	0.645±0.01	0.491±0.04	**0.707±0.01**
Average	0.399	0.337	0.448	0.426	0.299	**0.470**

STEM: corrected DiT (T=1000, 34M params).

Table 2. Cross-organ LOOO transfer, per-cohort Pearson ↑ (HEST-1k). Mean_{std} over three seeds; nested-validation. Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	STEM ICLR'25	LoGST+CONCH
IDC Breast	0.198±0.05	0.458±0.01	0.465±0.02	**0.529±0.02**	0.420±0.03	0.520±0.02
COAD Colorectal	0.420±0.06	0.670±0.03	0.669±0.03	0.662±0.01	0.452±0.02	**0.729±0.03**
CCRCC Kidney	0.068±0.01	0.055±0.01	0.106±0.01	0.095±0.01	0.090±0.01	**0.109±0.01**
LUNG Lung	0.493±0.07	0.378±0.18	0.657±0.01	0.624±0.01	0.512±0.03	**0.675±0.01**
PAAD Pancreas	0.459±0.02	0.399±0.02	0.463±0.00	0.537±0.02	0.409±0.03	**0.548±0.01**
PRAD Prostate	0.079±0.00	0.098±0.02	0.114±0.03	0.118±0.01	0.046±0.00	**0.118±0.01**
SKCM Skin	0.520±0.16	0.736±0.01	0.725±0.03	**0.748±0.01**	0.631±0.03	0.728±0.04
Average	0.319	0.399	0.457	0.473	0.366	**0.490**

STEM: corrected DiT (T=1000, 34M params).

random seeds (mean_{std}). Every model selects its checkpoint on an inner-validation split of the training fold and is evaluated once on the outer test-fold; where a cohort fold contains a single training slide, the inner split holds out a disjoint subset of that slide's spots. This removes the test-fold model selection present in earlier exploratory runs.

Baselines. We compare against ST-Net [3], Hist2ST [13], and HyperST [14]. All methods use identical frozen UNI features, data splits, and 50-gene panels; the prediction architecture is the principal varying factor. For reference, the published per-cohort accuracy of the higher-order method MCToGene [10] is summarized in Appendix D; it uses the ungrouped ten-cohort split and is not directly comparable to our feature-matched, nested-validation numbers.

Implementation. LoGST uses L=4 layers, hidden width d=256, 4 attention heads, k=8 neighbours, and 16 radial-basis functions (4.8M parameters). We train per slide with AdamW (learning rate 10^{-3}, weight decay 10^{-2}), a cosine schedule, and gradient clipping at 1.0, for at most 100 epochs with early-stopping patience of 20; each step samples at most 3,000 spots. Experiments run on one NVIDIA A100 GPU.

4.3. RQ1: Can simple direct-regression compete on the standard benchmark?

Answer: Yes—under controlled conditions LoGST leads all three baselines on every organ and metric. Under our feature-matched, nested-validation protocol, LoGST attains the best per-organ Pearson correlation on all seven retained HEST-1k organ groups and leads on every metric in the three-metric summary (Table 3: PCC 0.470, SCC 0.430, MSE 1.389; next-best HyperST PCC 0.448). The per-organ breakdown is in Table 1. For context, Appendix D reports published per-cohort numbers from the original HEST-1k split, including MCToGene [10]; those figures use ten cohorts without

Table 3. Intra-cohort three-metric summary (HEST-1k, averaged over seven organ groups; three seeds; UNI + CONCH features). PCC↑, SCC↑, MSE↓. Best per column in bold.

Method	PCC ↑	SCC ↑	MSE ↓
ST-Net [3] (NBE'20)	0.399	0.356	2.043
Hist2ST [13] (BIB'22)	0.337	0.321	1.616
HyperST [14] (CVPR'26)	0.448	0.411	1.439
BLEEP [12] (NeurIPS'23)	0.426	0.388	1.464
STEM [?] (ICLR'25)	0.299	0.274	1.855
LoGST (V3)	0.470	0.430	1.389

organ grouping and their original (not nested) selection protocol. On the standard split, larger and higher-order models score above a compact direct-regression model as expected; our finding is that under controlled conditions LoGST consistently leads the same three baselines while using one forward pass and 4.8M parameters.

4.4. RQ2: Does MorphoST generalize across organs without per-organ training?

Answer: Yes—LoGST is competitive under both LOOO and pooled transfer, on both benchmarks. We evaluate two cross-organ regimes on both benchmarks: leave-one-organ-out (looo), where the model trains on seven organ groups and is tested on the held-out eighth, and pooled five-fold cross-validation (pooled), where all organs are trained together. Table 4 summarizes pooled averages; Table 5 combines both regimes with PCC and MSE. LoGST achieves an average LOOO PCC of 0.421 on HEST-1k, leading all baselines; BLEEP is the closest at 0.416. LoGST leads on breast, colorectal, kidney, lung, pancreas, and prostate; HyperST leads on skin (0.720); Hist2ST leads on liver (0.033). In the pooled regime LoGST leads at 0.760, followed by BLEEP (0.758) and HyperST (0.756). To probe cross-species generalization, we additionally evaluate on 59 Visium slides spanning 7 mouse organs (brain, liver, skin, heart, bowel, kidney, muscle) from the MahmoodLab HEST collection [4]. Table 6 reports PCC under the same looo and pooled protocols using mouse-specific 50-gene HVG panels; LoGST achieves a pooled PCC of $0.887_{0.005}$ on mouse tissue, leading all three baselines (HyperST 0.865, Hist2ST 0.829, ST-Net 0.357).

Table 4. Cross-organ POOLED, average Pearson ↑ (HEST-1k). 5-fold pooled cross-validation; mean_{std} over three seeds; nested-validation. Best in bold.

Method	HEST-1k
ST-Net [3] (NBE'20)	$0.370_{0.01}$
Hist2ST [13] (BIB'22)	$0.737_{0.01}$
HyperST [14] (CVPR'26)	$0.756_{0.01}$
BLEEP [12] (NeurIPS'23)	$0.758_{0.01}$
STEM [?] (ICLR'25)	$0.643_{0.01}$
LoGST (V3)	$\mathbf{0.760_{0.01}}$

Table 5. Cross-organ generalization (three seeds, mean_{std}). looo: leave-one-organ-out transfer; pooled: pooled sample-level cross-validation. PCC↑ / MSE↓ on HEST-1k.

	looo		pooled	
Method	PCC	MSE	PCC	MSE
ST-Net [3] (NBE'20)	$0.267_{0.02}$	$1.846_{0.10}$	$0.370_{0.01}$	$0.907_{0.05}$
Hist2ST [13] (BIB'22)	$0.360_{0.02}$	$1.182_{0.07}$	$0.737_{0.01}$	$0.488_{0.01}$
HyperST [14] (CVPR'26)	$0.395_{0.01}$	$1.136_{0.10}$	$0.756_{0.01}$	$0.465_{0.02}$
BLEEP [12] (NeurIPS'23)	$0.416_{0.01}$	$\mathbf{1.010_{0.02}}$	$0.758_{0.01}$	$\mathbf{0.427_{0.02}}$
STEM [?] (ICLR'25)	$0.322_{0.01}$	$1.274_{0.00}$	$0.643_{0.01}$	$0.594_{0.01}$
LoGST (V3)	$\mathbf{0.421_{0.01}}$	$1.233_{0.11}$	$\mathbf{0.760_{0.01}}$	$0.440_{0.02}$

Table 6. Cross-organ generalization on mouse HEST (three seeds, mean_{std}). 59 Visium slides across 7 mouse organs (brain, liver, skin, heart, bowel, kidney, muscle). looo: leave-one-organ-out transfer; pooled: 5-fold pooled cross-validation. PCC↑ on 50 mouse HVGs. Same three feature-matched baselines as Table 1.

Method	looo PCC†	pooled PCC
ST-Net [3] (NBE'20)	$0.013_{0.011}$	$0.357_{0.036}$
Hist2ST [13] (BIB'22)	$0.018_{0.007}$	$0.829_{0.028}$
HyperST [14] (CVPR'26)	$\mathbf{0.019_{0.006}}$	$0.865_{0.013}$
BLEEP [12] (NeurIPS'23)	$-0.006_{0.003}$	$0.848_{0.012}$
STEM [?] (ICLR'25)	–	$0.279_{0.009}$
LoGST (V3)	$0.018_{0.006}$	$\mathbf{0.887_{0.005}}$

†All looo values are near floor due to cross-organ gene-panel mismatch (see text).

4.5. RQ3: Which spatial context components drive performance?

Answer: Global and slide context dominate; local attention is additive for intra-cohort but marginal for cross-organ transfer. Table 7 reports a factorial study that isolates local attention (L), global attention (G), and the slide representation (S) rather than adding them in a fixed sequence, evaluated in the cross-organ looo regime. Global and slide context contribute most: the strongest configuration is

Table 7. Factorial context ablation (intra-cohort, Pearson mean_{std} over three seeds on SKCM and LUNG). L, G, and S denote local attention, global attention, and slide representation. Local attention is the dominant contributor within an organ.

Variant	L	G	S	SKCM	LUNG	Avg
C000				$0.639_{0.00}$	$0.577_{0.01}$	0.608
C001			•	$0.633_{0.01}$	$0.578_{0.00}$	0.605
C010		•		$0.641_{0.01}$	$0.577_{0.00}$	0.609
C011		•	•	$0.639_{0.02}$	$0.583_{0.00}$	0.611
C100	•			$0.693_{0.01}$	$0.587_{0.00}$	0.640
C101	•		•	$0.695_{0.01}$	$0.585_{0.01}$	0.640
C110	•	•		$0.693_{0.01}$	$0.584_{0.00}$	0.638
C111	•	•	•	$\mathbf{0.696_{0.00}}$	$\mathbf{0.589_{0.01}}$	**0.643**

Table 8. Numerical coordinate-invariance check. Maximum absolute prediction difference when only the spot coordinates are transformed (randomly initialized model, 257-spot synthetic slide).

Transformation	Max. abs. difference
Translation	5.97×10^{-5}
Rotation	1.97×10^{-5}
Reflection	0
Spot permutation	5.96×10^{-7}

Table 9. Objective ablation (HEST-1k). Effect of adding the Pearson correlation term to the MSE objective, averaged over eight organ groups; three seeds.

Experiment	Setting	Pearson
HEST loss	LoGST, MSE only	0.382
	LoGST, MSE+PCC	0.389

global+slide (C011, 0.433), and every configuration containing G or S improves over the image-only baseline (C000, 0.417). Local attention is marginal for cross-organ transfer—adding it to the full model (C111, 0.421) does not improve over C011—although it is beneficial in the intra-cohort regime where the complete model is used. Table 9 separates the correlation-aware objective from the architecture: the correlation term raises both LoGST and the strongest baseline by a comparable margin (+0.007 and +0.010), so it does not account for the architecture gap, and LoGST transfers to the second benchmark (pooled PCC 0.652).

Table 8 verifies that the coordinate pathway is invariant to rigid-body transformations of the spot coordinates.

[UMAP of learned slide representations colored by organ — generated from the completed corrected checkpoints.]

Figure 3. Learned representation. UMAP of LoGST slide representations, colored by organ. Generated from the corrected checkpoints once training completes.

Table 10. Downstream spatial domain recovery. Agreement between spatial domains clustered from predicted vs. measured expression on held-out slides (higher is better). ARI/NMI over held-out slides; three seeds.

Method	ARI ↑	NMI ↑
ST-Net [3] (NBE'20)	$0.148_{0.13}$	–
Hist2ST [13] (BIB'22)	–	–
HyperST [14] (CVPR'26)	–	–
BLEEP [12] (NeurIPS'23)	–	–
STEM [?] (NeurIPS'24)	–	–
LoGST (V3)	$\mathbf{0.170_{0.15}}$	–

4.6. RQ4: Do predictions recover biological spatial structure beyond per-gene correlation?

Answer: Yes—predicted expression maps recover tissue spatial domains at parity with the strongest baseline. Scalar correlation measures per-gene accuracy but not whether predictions are spatially coherent. We probe this in two ways. First, Figure 3 visualizes LoGST slide representations colored by organ, checking whether the coordinate-invariant pathway yields organ-transferable features without supervision. Second, we cluster spots by predicted and by measured expression and compare the resulting spatial domains using ARI and NMI; high agreement means predictions recover tissue structure, not just per-gene trends (Table 10). All methods recover substantial structure (ARI well above chance), and LoGST matches the strongest baseline, confirming its accuracy advantage does not come at the expense of spatial coherence. Figure 4 gives a qualitative view: predicted expression maps for representative marker genes on held-out slides reproduce spatial gradients and tissue-structured patterns rather than a spatially uniform mean.

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[Predicted vs. measured spatial expression maps for marker genes on held-out slides — generated from the saved test predictions.]

Figure 4. Predicted expression maps. Measured (top) and LoGST-predicted (bottom) spatial expression of marker genes on held-out slides. Generated from the saved corrected test predictions.

4.7. RQ5: How efficient and scalable is the model?

Answer: LoGST uses 4.8M parameters and one forward pass, scaling to 10,000 spots in 65 ms on an A100. Table 11 compares parameter counts and inference procedures. LoGST contains 4.88M parameters, compared with 0.47M for ST-Net. It produces a slide-level prediction in one forward pass; its primary efficiency advantage is its simple inference procedure rather than the smallest parameter count. Table 12 reports measured model-only inference scaling.

Table 11. Model complexity. Parameter counts and inference mechanisms. LoGST predicts in a single forward pass. Maximum parameter count in bold.

Model	#Params (M)	Inference
ST-Net [3] (NBE'20)	0.47	single-pass
Hist2ST [13] (BIB'22)	2.18	single-pass
HyperST [14] (CVPR'26)	3.99	single-pass
BLEEP [12] (NeurIPS'23)	0.54	single-pass
STEM [?] (NeurIPS'24)	34.16	iterative
LoGST (V3)	4.88	single-pass

Table 12. LoGST inference scaling on an NVIDIA A100 80GB. Median model-only latency and peak allocated GPU memory over synthetic slides; feature extraction and disk I/O are excluded.

Spots	Latency (ms)	Memory (MiB)
500	3.48	43
1,000	3.63	67
3,000	9.36	335
5,000	21.31	842
10,000	64.74	3,183

5. Discussion

The experimental design tests two hypotheses: whether direct regression can attain strong accuracy under matched pathology features and splits, both within a cohort and under organ shift, and which combination of local, global, and slide-level context is most effective. Final conclusions will be drawn from the completed nested-validation runs. Independently of predictive accuracy, the architecture retains single-pass inference and coordinate-system invariance, and the representation and spatial domain analyses probe whether its predictions are structured rather than merely well correlated on average.

Limitations. First, LoGST uses a correlation-aware objective, whereas the regression baselines retain MSE; Table 9 isolates this factor, but a fully controlled comparison should evaluate all architectures under both losses. Second, all methods use frozen UNI embeddings and cannot recover information omitted by the feature encoder. Third, the reported experiments use a 50-gene prediction task; larger panels and additional platforms, staining variation, and institutions are needed to establish broader generalization. The coordinate-invariance result does not imply rotation invariance of the frozen histology encoder. Finally, the pooled cross-organ split is sample-disjoint but cannot be verified as patient-disjoint for cohorts without patient metadata.

6. Conclusion

We introduced LoGST, a direct-regression transformer for spatial gene expression prediction from histology. The model combines distance-biased local attention, global self-attention, and a slide-level representation; its coordinate pathway is invariant to rotations, reflections, and translations. Under a unified feature-matched protocol we evaluate LoGST in two regimes—intra-cohort and cross-organ transfer—on HEST-1k against direct, retrieval-based, and attention-based baselines, with three metrics, a factorial ablation, representation and spatial domain analyses, and measured scaling. Together these establish when direct spatial regression is a simple, reproducible, and generalizable alternative to generative prediction.

A. Full per-cohort intra-cohort results

Tables 13 and 14 give feature-matched intra-cohort comparisons for the seven retained cohorts (HCC/liver, LYMPH_IDC, and READ excluded; see Appendix E).

B. Cross-organ LOOO transfer using UNI

Table 15 gives the per-organ LOOO comparison using UNI features (1024-dim) with the same leave-one-organ-out protocol on HEST-1k.

C. Cross-organ LOOO transfer using UNI + CONCH

Table 16 gives the feature-matched UNI + CONCH LOOO comparison, with all three seeds complete for every method and organ.

D. Published Per-Cohort Comparison

E. HEST-1k Cohort Statistics

Table 18 lists the number of samples, total spots, and cross-validation folds for each HEST-1k cohort. Two organ groups in HEST-1k contain multiple cohorts: breast (IDC + LYMPH_IDC) and colorectal (COAD + READ). Within each group the constituent cohorts differ substantially in size: IDC has 2.25× more spots than LYMPH_IDC (44,974 vs. 19,964), and COAD has 2.2× more spots than READ (18,523 vs. 8,407). To avoid misrepresentation from unweighted averaging over unequal cohorts, Table 1 uses IDC as the representative breast cohort and COAD as the representative colorectal cohort, discarding the smaller partner cohorts (LYMPH_IDC and READ) from the organ-level rows. This choice maximises statistical reliability of each organ-level estimate.

References

[1] Richard J Chen, Tong Ding, Ming Y Lu, et al. Towards a general-purpose foundation model for computational pathology. Nature Medicine, 30:850–862, 2024. 2, 4

[2] Youngmin Chung, Ji Hun Ha, Kyeong Chan Im, and Joo Sang Lee. Accurate spatial gene expression prediction by integrating multi-resolution features. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. 2

[3] Bryan He, Ludvig Bergenstråhle, Linnea Stenbeck, et al. Integrating spatial gene expression and breast tumour morphology via deep learning. Nature Biomedical Engineering, 4(8):827–834, 2020. 2, 5, 6, 7, 8, 12

[4] Guillaume Jaume, Paul Doucet, Andrew H Song, et al. HEST-1k: A dataset for spatial transcriptomics and histology image analysis. In Advances in Neural Information Processing Systems (NeurIPS), 2024. 3, 4, 6

[5] others. A large-scale benchmark of cross-modal learning for histology and gene expression in spatial transcriptomics. arXiv preprint arXiv:2508.01490, 2025. 3

[6] Minxing Pang, Kenong Su, and Mingyao Li. Leveraging information in spatial transcriptomics to predict super-resolution gene expression from histology images in tumors. bioRxiv, 2021. 2

[7] Victor Garcia Satorras, Emiel Hoogeboom, and Max Welling. E(n) equivariant graph neural networks. In International Conference on Machine Learning (ICML), 2021. 2

[8] Yaxuan Song, Jianan Fan, Hang Chang, and Weidong Cai. Gene-dml: Dual-pathway multi-level discrimination for gene expression prediction from histopathology images. In IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2026. 2

[9] Patrik L Ståhl, Fredrik Salmén, Sanja Vickovic, et al. Visualization and analysis of gene expression in tissue sections by spatial transcriptomics. Science, 353 (6294):78–82, 2016. 1

[10] Youhan Sun, Jiahua Rao, Kangrui Du, Jiancong Xie, and Yuedong Yang. Predicting spatial transcriptomics from histology images via high-order multi-cell interaction modeling. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. 2, 5, 12

[11] Ashish Vaswani, Noam Shazeer, Niki Parmar, et al. Attention is all you need. In Advances in Neural Information Processing Systems (NeurIPS), 2017. 4

[12] Ronald Xie, Kuan Pang, Sai Chung, et al. Spatially resolved gene expression prediction from histology images via bi-modal contrastive learning. In Advances in Neural Information Processing Systems (NeurIPS), 2023. 2, 6, 7, 8

[13] Yuansong Zeng, Zhuoyi Wei, Weijiang Yu, et al. Spatial transcriptomics prediction from histology jointly through transformer and graph neural networks. Briefings in Bioinformatics, 23(5), 2022. 2, 5, 6, 7, 8, 12

[14] Chen Zhang, Yilu An, Ying Chen, Hao Li, Xitong Ling, Lihao Liu, Junjun He, Yuxiang Lin, Zihui Wang, and Rongshan Yu. Hyperst: Hierarchical hyperbolic learning for spatial transcriptomics prediction. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026. 2, 5, 6, 7, 8, 12

[15] Xin Zhang et al. Stpath: a generative foundation model for integrating spatial transcriptomics and whole-slide images. npj Digital Medicine, 2025. 2

Table 13. Intra-cohort results using UNI features (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI features (1024-dim) and 50-gene panels. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix E. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	DeepSpaCE Sci Rep'22	Gene-DML WACV'26	TRIPLEX CVPR'24	LoGST (V3)
IDC Breast	0.538±0.01	0.250±0.07	0.558±0.01	0.508±0.01	0.506±0.07	0.532±0.01	0.453±0.02	0.501±0.03	0.557±0.01	0.555±0.01	**0.567±0.00**
COAD Colorectal	0.209±0.02	0.293±0.01	0.279±0.01	0.308±0.01	**0.323±0.01**	0.274±0.00	0.198±0.01	0.266±0.01	0.276±0.01	0.293±0.00	0.299±0.00
CCRCC Kidney	0.172±0.01	0.217±0.01	0.256±0.00	0.184±0.01	0.253±0.01	0.237±0.01	0.134†	0.234±0.01	0.224±0.03	0.228±0.03	**0.265±0.02**
LUNG Lung	0.482±0.01	0.378±0.10	0.581±0.01	0.559±0.01	0.579±0.00	0.561±0.00	0.463±0.01	0.546±0.00	0.564±0.00	0.569±0.00	**0.589±0.01**
PAAD Pancreas	0.353±0.02	0.243±0.06	0.367±0.04	**0.393±0.03**	0.361±0.03	0.296±0.05	0.292±0.02	0.290±0.04	0.362±0.04	0.362±0.03	0.392±0.04
PRAD Prostate	0.251±0.01	0.317±0.02	0.369±0.00	0.343±0.01	**0.400±0.00**	0.370±0.02	0.215±0.01	0.391±0.00	0.363±0.01	0.370±0.01	0.390±0.02
SKCM Skin	0.632±0.01	0.509±0.02	0.641±0.00	0.639±0.01	0.656±0.02	0.610±0.01	0.559±0.01	0.628±0.00	0.674±0.00	0.647±0.00	**0.696±0.00**
Average	0.377	0.315	0.436	0.419	0.440	0.411	0.330†	0.408	0.432	0.432	**0.457**

†STEM CCRCC has one completed seed; all other entries use three seeds.

Table 14. Intra-cohort results using UNI + CONCH features (Pearson mean_{std} over three seeds; nested-validation). All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels. HCC (liver), LYMPH_IDC, and READ are excluded per Appendix E. Best per row in bold.

Cohort	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	LoGST+CONCH
IDC Breast	0.551±0.01	0.309±0.06	0.567±0.01	0.519±0.00	0.571±0.01	0.551±0.01	0.439±0.01	**0.581±0.01**
COAD Colorectal	0.242±0.01	0.295±0.01	0.285±0.01	0.314±0.01	**0.328±0.01**	0.293±0.01	0.177±0.01	0.318±0.01
CCRCC Kidney	0.163±0.02	0.184±0.02	**0.256±0.02**	0.175±0.02	0.210±0.02	0.223±0.02	0.123±0.00	0.252±0.01
LUNG Lung	0.518±0.01	0.470±0.01	0.587±0.00	0.562±0.01	**0.598±0.00**	0.584±0.01	0.423±0.03	0.589±0.01
PAAD Pancreas	0.359±0.03	0.247±0.02	0.404±0.04	0.419±0.03	0.413±0.04	0.415±0.03	0.233±0.02	**0.437±0.03**
PRAD Prostate	0.294±0.02	0.333±0.02	0.384±0.02	0.348±0.01	**0.413±0.01**	0.377±0.03	0.207±0.02	0.410±0.01
SKCM Skin	0.665±0.01	0.522±0.03	0.653±0.01	0.645±0.01	0.654±0.01	0.631±0.01	0.491±0.04	**0.707±0.01**
Average	0.399	0.337	0.448	0.426	0.455	0.439	0.299	**0.470**

All entries use three completed seeds. STEM uses the corrected DiT configuration ($T = 1000$, 100 epochs).

Table 15. Cross-organ LOOO transfer using UNI features (Pearson mean_{std} over three seeds; nested-validation; liver excluded). Leave-one-organ-out: trained on all other organs, tested on the held-out organ. All methods use identical frozen UNI features (1024-dim) and 50-gene panels. Best per row in bold.

Organ	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	Gene-DML WACV'26	TRIPLEX CVPR'24	LoGST (V3)
IDC Breast	0.217±0.06	0.459±0.02	0.501±0.02	0.509±0.03	0.452±0.01	0.531±0.03	0.406±0.01	0.521±0.03	0.528±0.02	**0.534±0.02**
COAD Colorectal	0.430±0.03	0.688±0.04	0.669±0.06	0.667±0.02	0.672±0.03	**0.749±0.01**	0.504±0.03	0.737±0.01	0.726±0.03	0.703±0.01
CCRCC Kidney	0.084±0.03	0.084±0.01	0.108±0.01	0.100±0.02	0.122±0.01	0.106±0.01	0.090±0.01	0.116±0.02	0.113±0.01	**0.124±0.00**
LUNG Lung	0.504±0.04	0.499±0.06	0.650±0.01	0.621±0.02	0.667±0.01	0.666±0.01	0.561±0.01	0.676±0.01	**0.680±0.01**	0.660±0.01
PAAD Pancreas	0.404±0.05	0.360±0.07	0.397±0.03	0.521±0.02	0.522±0.01	**0.551±0.02**	0.460±0.02	0.526±0.03	0.521±0.02	0.507±0.04
PRAD Prostate	0.071±0.01	0.104±0.02	0.092±0.03	0.111±0.00	0.107±0.02	0.098±0.02	0.068±0.01	0.110±0.02	0.112±0.03	**0.119±0.02**
SKCM Skin	0.402±0.08	0.652±0.08	0.720±0.02	0.745±0.01	0.724±0.02	0.749±0.01	0.686†	**0.752±0.02**	0.742±0.03	0.692±0.06
Average	0.302	0.407	0.448	0.468	0.466	**0.493**	0.396†	0.491	0.489	0.477

†STEM uses three seeds for all organs except SKCM (one completed seed); its average is marked accordingly.

Table 16. Cross-organ LOOO transfer using UNI + CONCH features (Pearson mean_{std} over three seeds; nested-validation; liver excluded). Leave-one-organ-out: trained on all other organ groups, tested on the held-out cohort. All methods use identical frozen UNI + CONCH features (1536-dim) and 50-gene panels, including STEM's corrected DiT configuration ($T = 1000$, 100 epochs). Best per row in bold.

Organ	ST-Net NBE'20	Hist2ST BIB'22	HyperST CVPR'26	BLEEP NeurIPS'23	M2OST AAAI'25	EGN WACV'23	STEM ICLR'25	LoGST+CONCH
IDC Breast	0.198±0.05	0.458±0.01	0.465±0.02	0.529±0.02	0.437±0.04	**0.567±0.02**	0.420±0.03	0.520±0.02
COAD Colorectal	0.420±0.06	0.670±0.03	0.669±0.03	0.662±0.01	0.657±0.05	0.728±0.01	0.452±0.02	**0.729±0.03**
CCRCC Kidney	0.068±0.01	0.055±0.01	0.106±0.01	0.095±0.01	0.081±0.03	0.103±0.02	0.090±0.01	**0.109±0.01**
LUNG Lung	0.493±0.07	0.378±0.18	0.657±0.01	0.624±0.01	0.665±0.01	0.672±0.01	0.512±0.03	**0.675±0.01**
PAAD Pancreas	0.459±0.02	0.399±0.02	0.463±0.00	0.537±0.02	0.513±0.03	**0.572±0.01**	0.409±0.03	0.548±0.01
PRAD Prostate	0.079±0.00	0.098±0.02	0.114±0.03	0.118±0.01	0.087±0.02	0.107±0.01	0.046±0.00	**0.118±0.01**
SKCM Skin	0.520±0.16	0.736±0.01	0.725±0.03	0.748±0.01	0.740±0.01	**0.750±0.02**	0.631±0.03	0.728±0.04
Average	0.319	0.399	0.457	0.473	0.454	**0.500**	0.366	0.490

STEM values are mean_{std} over three completed seeds.

Table 17. Published per-cohort accuracy on the standard HEST-1k splits, mean PCC↑ as reported in respective papers (ten cohorts, no organ grouping). Shown for context only—not comparable to Table 1, which uses organ grouping and nested-validation. †LoGST uses our organ-grouped splits, not the standard per-cohort split.

Method	HEST-1k (10 cohorts)
ST-Net [3] (NBE'20)	0.286
Hist2ST [13] (BIB'22)	running
HyperST [14] (CVPR'26)	running
MCToGene [10] (CVPR'26)	**0.435**
LoGST† (ours)	0.440

Table 18. HEST-1k cohort statistics. Spots counted after barcode filtering. Folds = number of train/test splits in the per-cohort cross-validation. HCC (liver) is excluded from Table 1 due to its near-zero signal floor.

Cohort	Organ	Samples	Spots	Folds
IDC	Breast	4	44,974	4
LYMPH_IDC	Breast	4	19,964	4
COAD	Colorectal	4	18,523	2
READ	Colorectal	4	8,407	2
CCRCC	Kidney	24	74,220	6
HCC	Liver*	2	4,248	2
LUNG	Lung	2	7,505	2
PAAD	Pancreas	3	9,793	3
PRAD	Prostate	23	62,710	2
SKCM	Skin	2	5,716	2

*HCC (liver) is included here for completeness but excluded from Table 1; all baselines score near zero on this cohort due to low spatial gene-expression variance.